"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all versions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues"

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1.

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2. Since the vector space with vectors satisfying $x - 5y = 0$, is of dimension 4, R^n must be of dimension $n \geq 5$, such tha $x \subseteq R^5$.

Then $K = \{(x,y) \in R^5 \mid x - 5y = 0\} \subseteq R^5$

4. $$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix}$$

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$$= I^2 - I.2uu^T - 2.uu^T.I + 4uu^T.uu^T$$

$$= I.(I - 2.u.u^T - 2.u.u^T) + 4u.\|u\|^2.u^T$$

$$= I - 2.u.u^T - 2.u.u^T + 4u.\|u\|^2.u^T$$

$$\|u\|^2 = u^T.u$$

$$\|u\| = \pm\sqrt{u^T.u} = \|u\| = 1$$

Thus; $H^2 = I - 4.u.u^T + 4.u.u^T$ $\boxed{H^2 = I}$

$$H^2 = I$$

b) $H.H^T = (I - 2u.u^T)(I - 2.u.u^T)^T$

$$= (I - 2.u.u^T)(I^T - (2.u.u^T)^T)$$

$$= (I - 2.u.u^T)(I - 2(u^T)^T u^T)$$

$$= (I - 2.u.u^T)(I - 2.uu^T)$$

$$= I^2 - 2uu^T - 2.u.u^T + 4.u.u^T.u.u^T$$

$$= I - 4uu^T + 4.u\|u\|^2 u^T$$

$$= I - 4uu^T + 4u.u^T = I = H.H^T$$

Thus; $H^T.H = I$

Thus; $H.H^T = H^T H = I$

Thus H is an orthogonal matrix

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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[illegible] a matrix K, a vector will be eigen-vector of K corresponding eigenvalue λ if $K.V = \lambda V$

$$HU = (I - 2U.U^T).U = IU - 2U.U^TU$$
$$= U - 2U.\|U\|^2$$

(As I is identity matrix. Therefore 1 is the only eigenvalue of I, and $I.U = 1.U = U$)

and $\|U\| = +\sqrt{U^TU}$ and U is a unit vector, thus $\|U\| = 1 = +\sqrt{U^T.U}$ and thus $U^T.U = \|U\|^2 = 1$ thus

$$HU = U - 2.U.1 = -U$$
$$HU = -U = (-1).U$$
$$HU = \lambda U, \quad \lambda = -1$$

thus U is an eigen vector of H and corresponding eigenvalue for U is $= \lambda = -1$

d) $H = (I - 2.U.U^T)$

$$H^T = (I - 2.U.U^T)^T$$
$$= I^T - 2.(U.U^T)^T$$
$$= I - 2.(U^T)^T.U^T$$
$$= I - 2.U.U^T = H$$

Therefore H is a symmetric matrix.

The theorem which shows;

$$H = Q\Lambda Q^T$$
$$= Q\Lambda Q^{-1}$$
$$= (Q^{-1})^{-1}\Lambda Q^{-1}$$
$$= P^{-1}\Lambda P$$

P is an orthogonal matrix

$Q.Q^T = I$

$Q^T = Q^{-1}$

Λ = orthogonal matrix consist of eigen values of H

Thus H is a diagonalizible matrix.